

SIVASANKARAMALAN GUNASEKARASIVAM

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🌐 [Portfolio](#)

10+

AI AGENTS BUILT

400+

HRS SAVED / QUARTER

90%

FASTER TEST CREATION

30+

ENGINEERS TRAINED

📄 PROFESSIONAL SUMMARY

- AI-native Quality Engineering leader with 11 years of experience building autonomous test agents, designing MCP-based tool orchestration, AI-assisted testing across enterprise engineering teams
- Engineered production-grade AI agents using GitHub Copilot that drove measurable impact across 7 teams, achieving 92% reduction in test case creation time while maintaining less than 1% error rate and 100% compliance with team quality standards
- Led team-wide AI testing transformation for the 30-person QE team, defining the AI testing roadmap, establishing AI adoption governance frameworks, and aligning AI-native workflows with sprint delivery cycles and product velocity goals
- Fostered AI innovation culture by mentoring 30+ engineers to independently build and ship their own AI tools, earning first-place recognition at the QA Hub Global Hackathon among 150+ participants
- Microsoft Certified GitHub Copilot Engineer with deep expertise in prompt engineering, multi-agent orchestration, and shift-left AI practices that bridge the gap between development and testing at the story level

🔧 CORE SKILLS

AI Agent Engineering Autonomous Test Agents, LLM Tool-Calling Loops, Multi-Agent Orchestration, Planner-Executor-Verifier Patterns, Self-Healing Test Pipelines

MCP Protocol and Tool Orchestration Playwright MCP, Appium MCP, Atlassian MCP, qTest MCP, Custom MCP Server Design, MCP-Compliant Tool Abstractions, Context Window Management

Prompt Engineering Multi-Shot Prompting, Role-Play Prompting, Temperature Tuning, Deterministic Test Output Generation, Prompt Validation Frameworks

AI-Augmented Test Design LLM-Generated Test Cases from User Stories, AI-Assisted Regression Prioritization, Semantic Test Matching, Flaky Test Pattern Detection, AI Coverage Gap Analysis

QE Fundamentals Shift-Left Testing, Risk-Based Test Strategies, API Testing, Contract Testing, Performance Testing, SDLC, Test Metrics, Flakiness Rate

CI/CD and DevOps Integration Jenkins, GitHub Actions, GitLab CI, Azure DevOps, Quality Gates, Pipeline Automation, Continuous Testing

People Leadership and AI Adoption Team of 24, AI Tool Training, AI Roadmap Definition, Cross-Team Adoption, Stakeholder Communication, Quality Gate Governance

🔧 TECHNICAL SKILLS

GitHub Copilot

Claude Code

Cursor

MCP Protocol

Playwright MCP

Appium MCP

Atlassian Rovo

Deepeval

Ragas

LangChain

LangGraph

LangSmith

OpenAI API

Anthropic API

Token Optimization

Prompt Engineering

LLM Tool-Calling

Java

Kotlin

Selenium WebDriver

Playwright

Appium

REST Assured

Cucumber

TestNG

JUnit

Postman

BDD

k6

JMeter

Jenkins

GitHub Actions

GitLab CI

Azure DevOps

Docker

Kubernetes

AWS

GCP

SQL

Git

Agile

Scrum

📁 PROFESSIONAL EXPERIENCE

Lead QA Engineer (Domain Lead - QE)

Oct 2023 - Present

EPAM Systems | India

- Built 10+ production-grade AI agents using GitHub Copilot and MCP protocol that collectively saved 400+ hours per quarter across 7 DCX teams, transforming manual QE workflows into automated intelligent systems with direct client approval and testimonials
- Designed and deployed a Test Case Creation Agent that generates qTest-compliant test cases from JIRA user stories through a single Copilot command, reducing creation time by 92% (1 hour to 30 minutes per story) while achieving less than 1% error rate across 100+ stories per month
- Built a Requirement Mapping Agent that automatically links JIRA requirements to qTest test cases via API integration, achieving 100% traceability with zero linking errors and 90% time reduction (5 minutes to 30 seconds per requirement)
- Engineered a Serenity Report Triage Agent that parses Jenkins artifacts, detects recurring failure patterns and flaky tests, performs stability trend analysis, and auto-generates daily Critical Path failure summaries, saving 35 hours per month across the QE team
- Created an AI Test Finder and Predictor Agent that semantically matches acceptance criteria, GWT statements, and API endpoints against existing test repositories, auto-commenting coverage suggestions on JIRA tickets to enable shift-left testing with developers
- Refactored instruction-based agents to enterprise-grade MCP architecture with structured tool orchestration, centralized governance, and multi-agent scalability, deployed across all DCX teams for platform-level standardization
- Extended QE-Copilot ecosystem with a Jenkins Trigger Agent enabling IDE-based job execution for CP, Regression, and Smoke suites, saving 49 hours per month across 7 engineers through standardized command-based triggering
- Led AI testing transformation for 30+ QE engineers, training the team on GitHub Copilot, Claude Code, MCP protocol, and Cursor while defining and executing the AI testing roadmap aligned with product velocity

- Mastered advanced prompt engineering including multi-shot prompting, role-play techniques, and temperature tuning to achieve deterministic test outputs with 100% compliance to team quality standards
- Established quality metrics and AI-driven test reporting dashboards covering automation ROI, agent adoption rates, and hours saved, delivering structured reports to engineering leadership and client stakeholders

Senior SDET

Apr 2022 - Oct 2023

Navi Technologies | Bangalore, India

- Designed end-to-end quality processes and test strategies for fintech microservices covering functional, regression, and integration testing, reducing regression time by 40% while establishing Agile-aligned release criteria and defect tracking practices across squads
- Built automation frameworks using Karate and REST Assured, automating 120+ critical workflows across payment flows and financial data validation, integrated into CI/CD pipelines providing continuous defect feedback to development teams
- Collaborated with developers, product managers, and business analysts to define quality standards, document and track defects, set release readiness gates, and drive continuous improvement in QA practices across delivery teams

Associate Tech Lead (Quality and Automation)

Apr 2020 - Apr 2022

OkCredit | Bangalore, India

- Owned quality assurance for a SaaS product serving 1M+ users, designing test plans and executing functional and non-functional tests to ensure product quality aligned with user expectations, scaling automation coverage from 60% to 95%
- Built reusable Selenium and Appium automation frameworks covering web, Android, and iOS platforms integrated into Jenkins CI/CD pipelines, enabling continuous automated testing on every deployment and reducing manual regression effort by 70%
- Mentored and coached a QA team on test automation, defect management, and quality best practices.

Senior Software Engineer (Quality - OTT and Cross-Device)

Nov 2019 - Apr 2020

Rakuten Viki | Singapore

- Led quality assurance for multi-device OTT platforms across 50+ configurations (iOS, Android, FireTV, AppleTV) in Singapore, conducting functional, performance, and network condition testing that reduced consumer-facing defects by 35%

Software Engineer

Jun 2017 - Oct 2019

Altisource | Bangalore, India

- Built Appium automation frameworks achieving 92% coverage across web workflows, integrated into Jenkins CI/CD pipelines reducing test execution time by 50% through parallel execution

Software Test Engineer

May 2015 - May 2017

AB Innovative | Bangalore, India

- Established test case management and defect tracking processes, creating reusable test libraries for common business workflows

AI AGENT PROJECTS

QE-Copilot Agent Ecosystem (10+ Agents)

- Architected a unified AI agent ecosystem within GitHub Copilot that spans the full QE lifecycle from test case generation through requirement mapping, test discovery, CI/CD triggering, and failure triage, used across 7 enterprise teams
- Delivered measurable ROI across 7 enterprise teams, with 100% requirement traceability and automated daily failure analysis replacing manual triage across all regression cycles

MCP Architecture Migration

- Migrated instruction-based Copilot agents to enterprise MCP architecture with structured tool abstractions, governance controls, and multi-agent orchestration support, enabling platform-level scalability across DCX teams

KEY ACHIEVEMENTS

- ◆ Led a GitHub Copilot hackathon for 300+ engineers at Signify in collaboration with GitHub and Microsoft teams, presenting best practices on GitHub Copilot adoption and building custom MCP servers for internal tooling
- ◆ Won first place at the QA Hub Global Hackathon among 150+ participants by delivering a complete SDLC agentic orchestration pipeline covering requirements, test generation, execution, and deployment
- ◆ Led AI adoption for the 30-person QE team on GitHub Copilot, Claude Code, MCP, and Cursor, earning direct client approval and testimonials from QE leadership for measurable quality improvement

CERTIFICATIONS

Microsoft Certified: GitHub Copilot Engineer - Microsoft

Professional Scrum Master - Scrum.org

EDUCATION

Executive MBA - Product Engineering | Indian Institute of Technology Madras (IIT Madras)

Bachelor of Engineering - Mechanical Engineering | Anna University